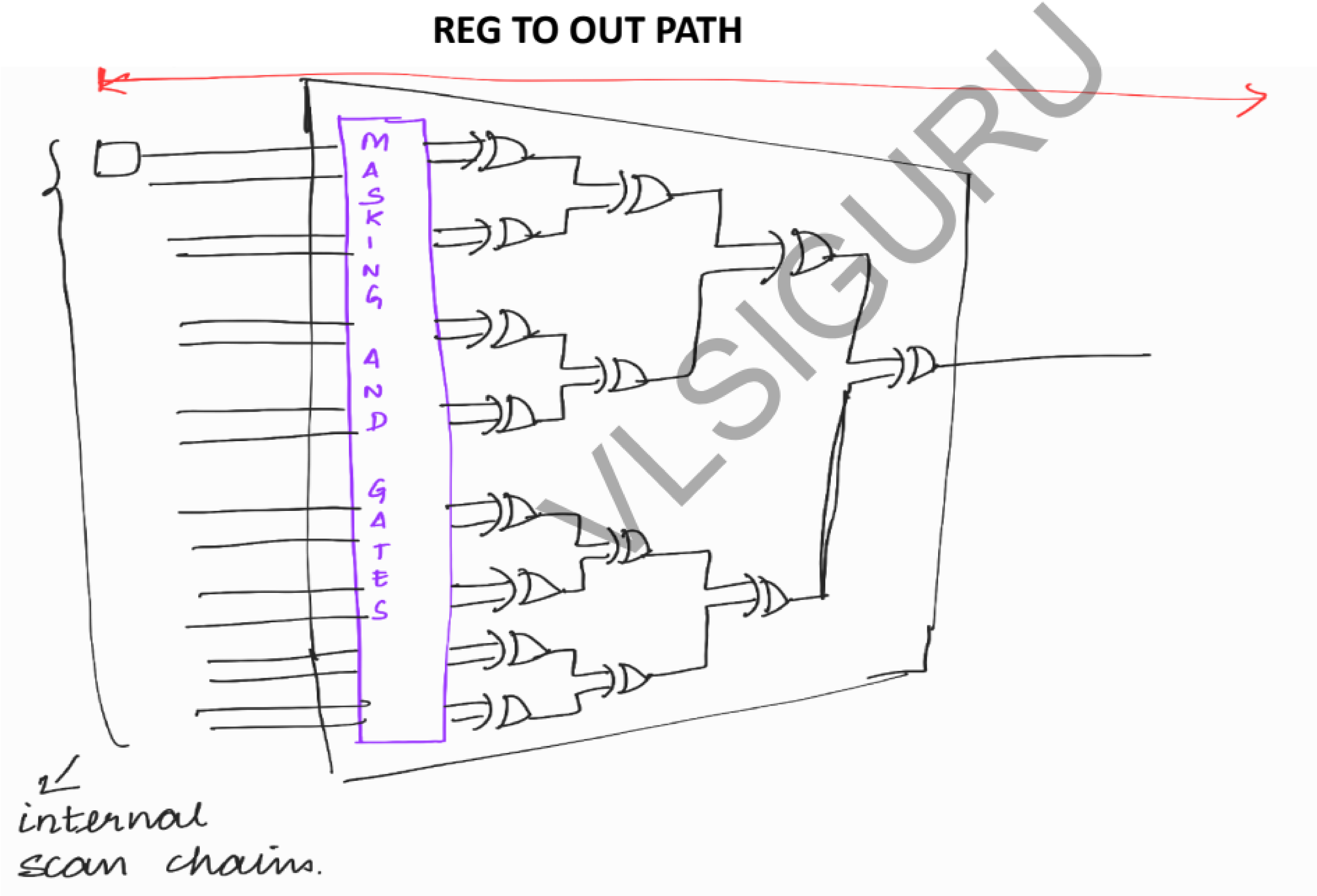
SCAN COMPRESSION (PART4)

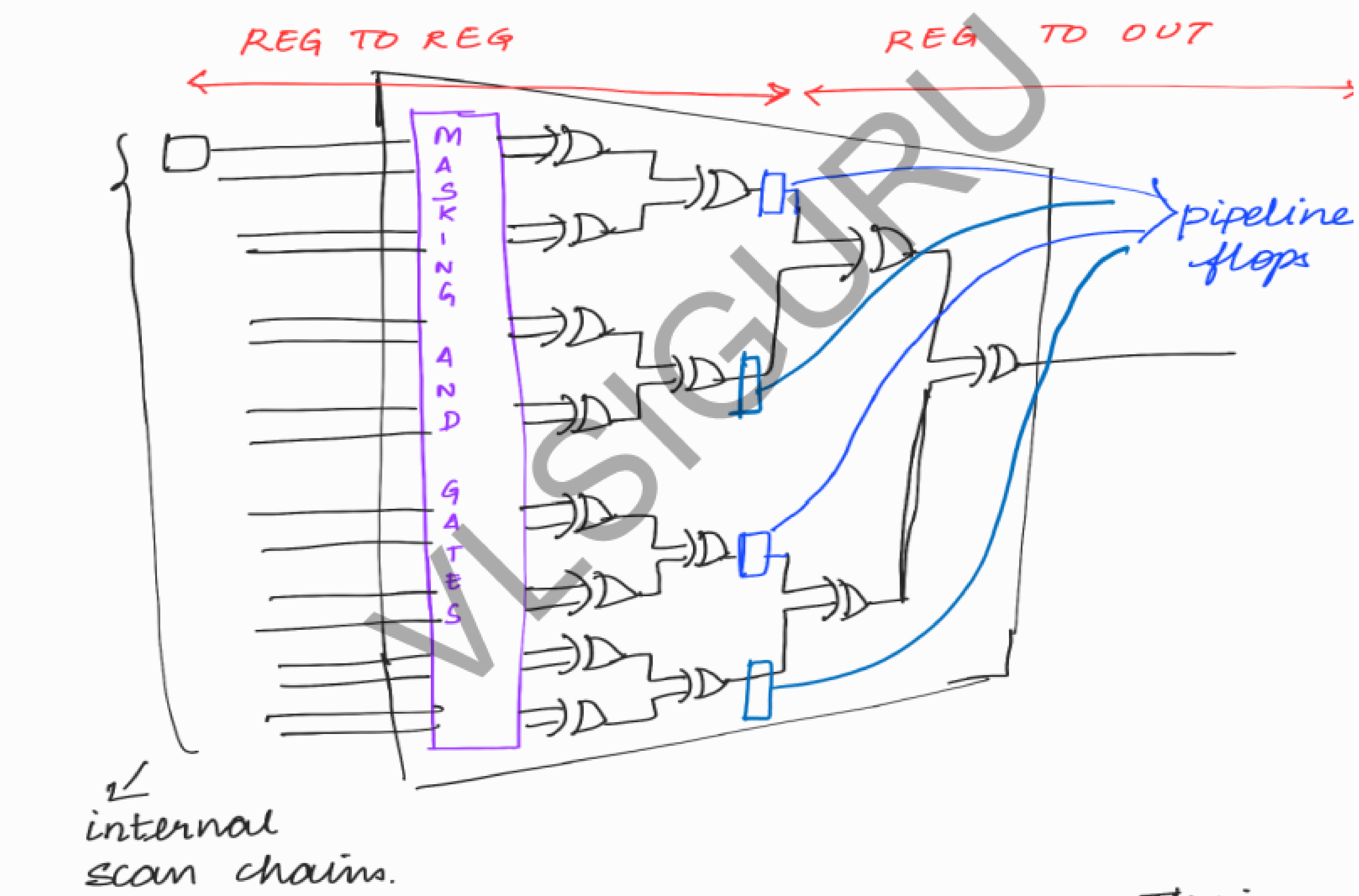
PIPELINE FLOPS IN COMPRESSOR LOGIC

If there are many EX-OR gates present inside the compressor, it is very difficult to meet the timing.



PIPELINE FLOPS IN COMPRESSOR LOGIC (Contd…)

In order to easily meet the timing, pipeline flops are added so that the big reg to out path will be broken into 2 paths – reg to reg and reg to out

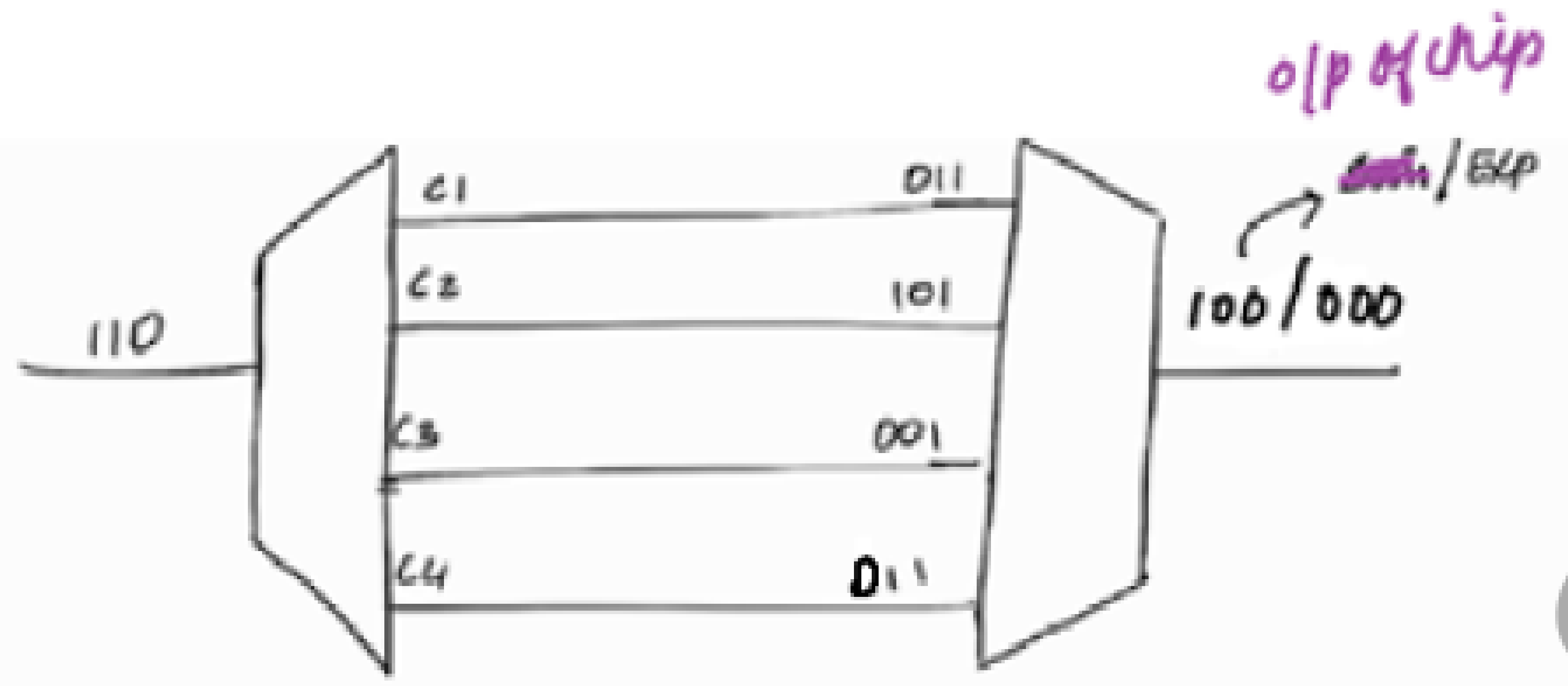


pipeline FFs are also not scan ffs. Their clock is EDT clock.

DEBUG OF EDT PATTERNS

EXAMPLE

WITH COMPRESSION



Failure log:-

~~Simulated Value~~ o/p of chip	Expected Value	channel no	cycle no	Pattern no.
1	0	1	3	1

Channel 1 drives chain 1, chain 2, chain 3 and chain 4.
Cycle 3 will come for chain 1, chain 2, chain 3 and chain 4.
But we exactly don't know in which chain mismatch has occurred.
So we go for 1-hot patterns.

DEBUG OF EDT PATTERNS (Contd…)

1-hot patterns

If we give information about which pattern is failing to the tool, the tool will generate 1-hot patterns for that particular failing pattern.

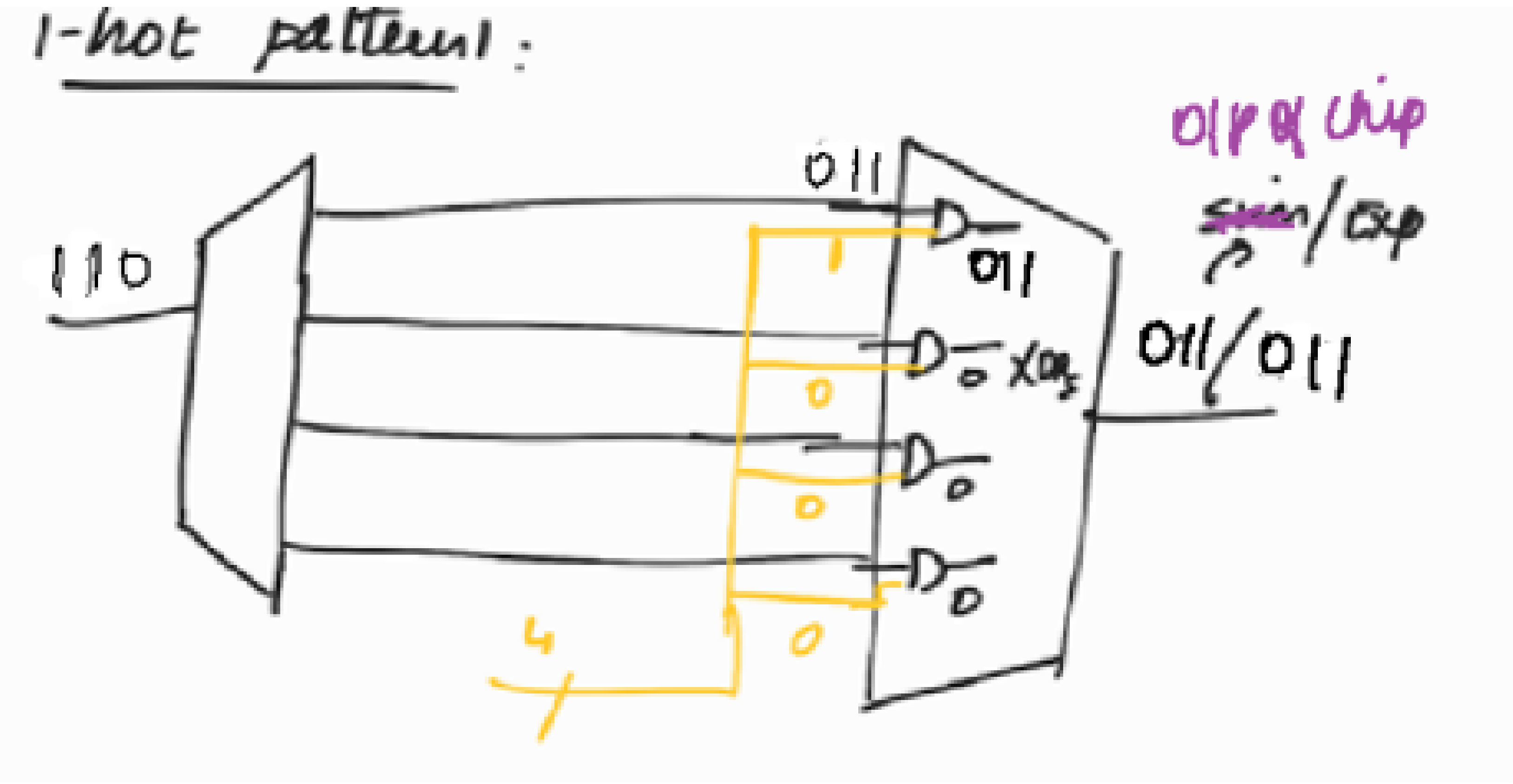
```
read_patterns production_edt_pat.stil
set_pattern_filtering –external -list 57 58 65
expand_compressed_patterns
```

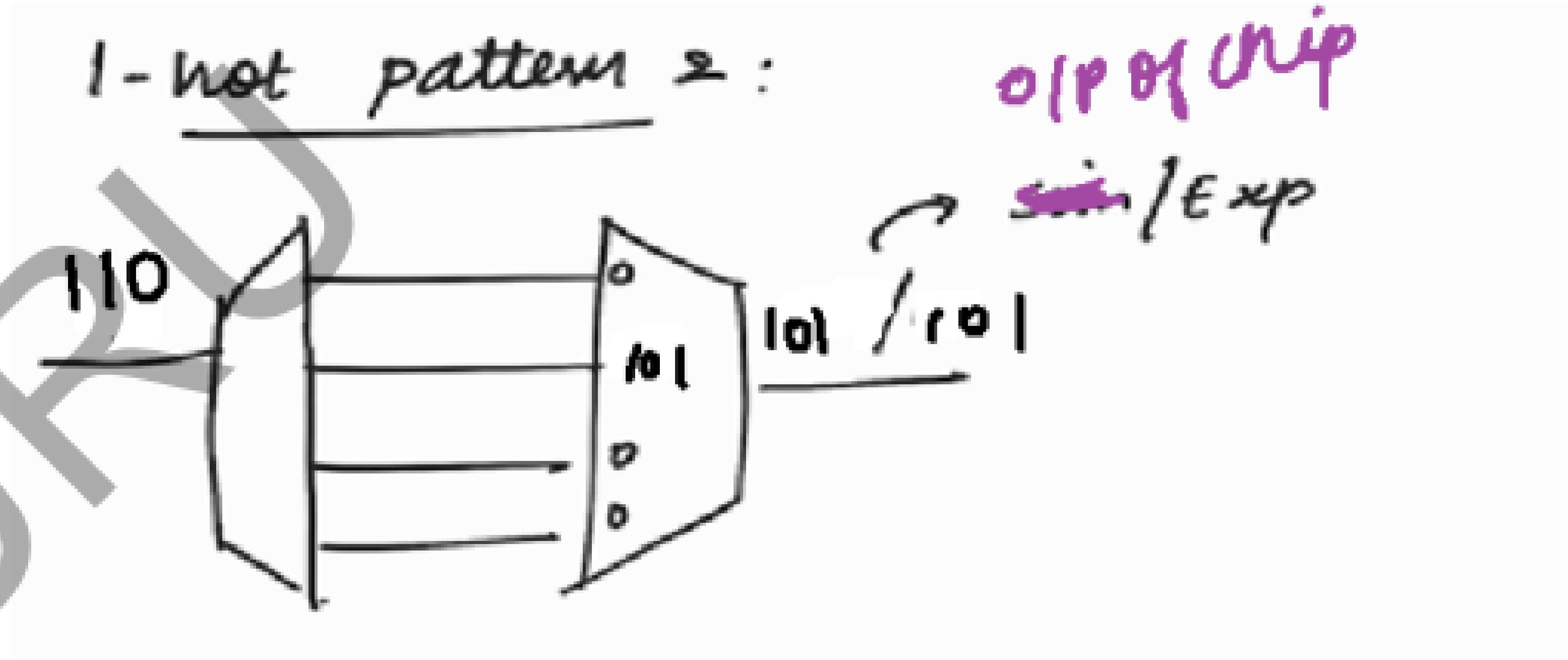
In 1-hot pattern, for each pattern only 1 chain will be enabled in a channel.

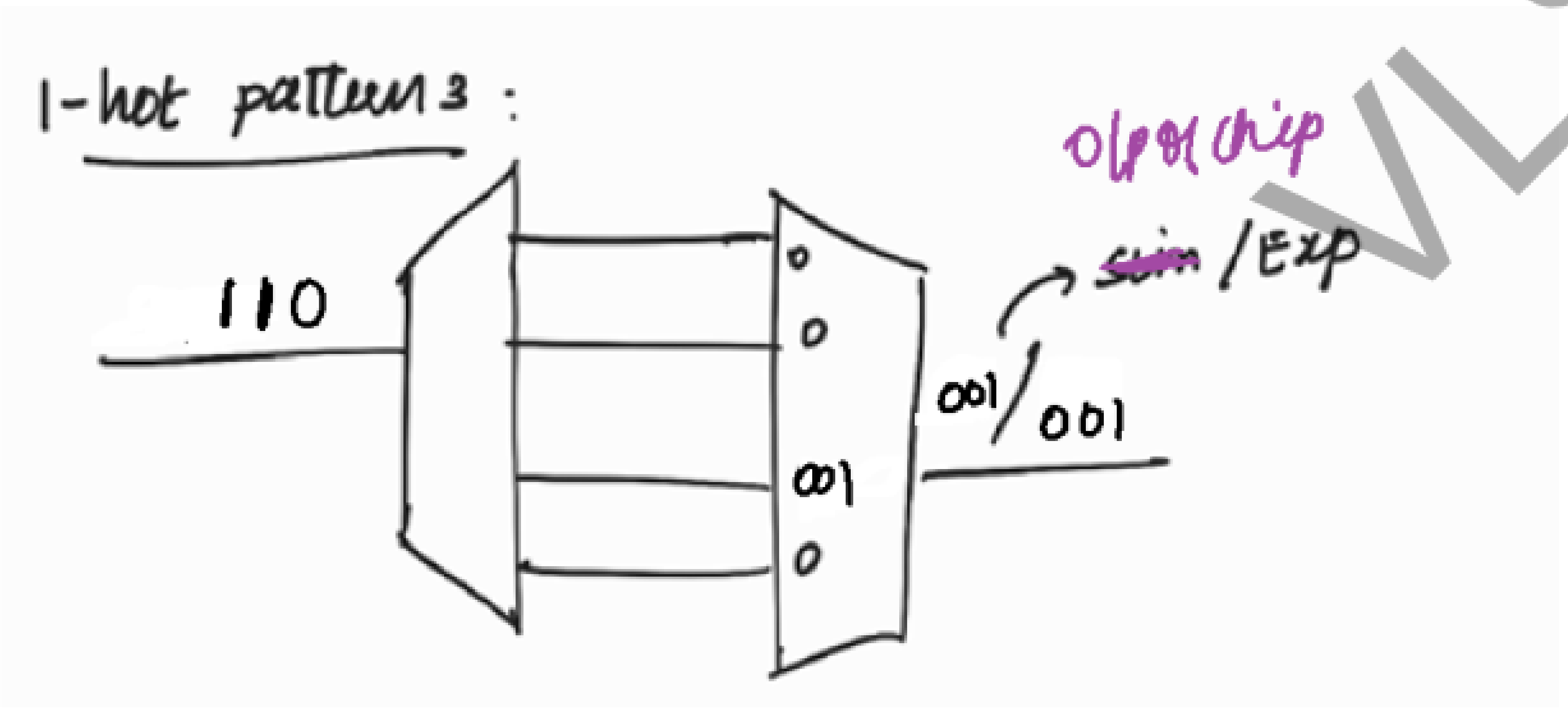
DEBUG OF EDT PATTERNS (Contd…)

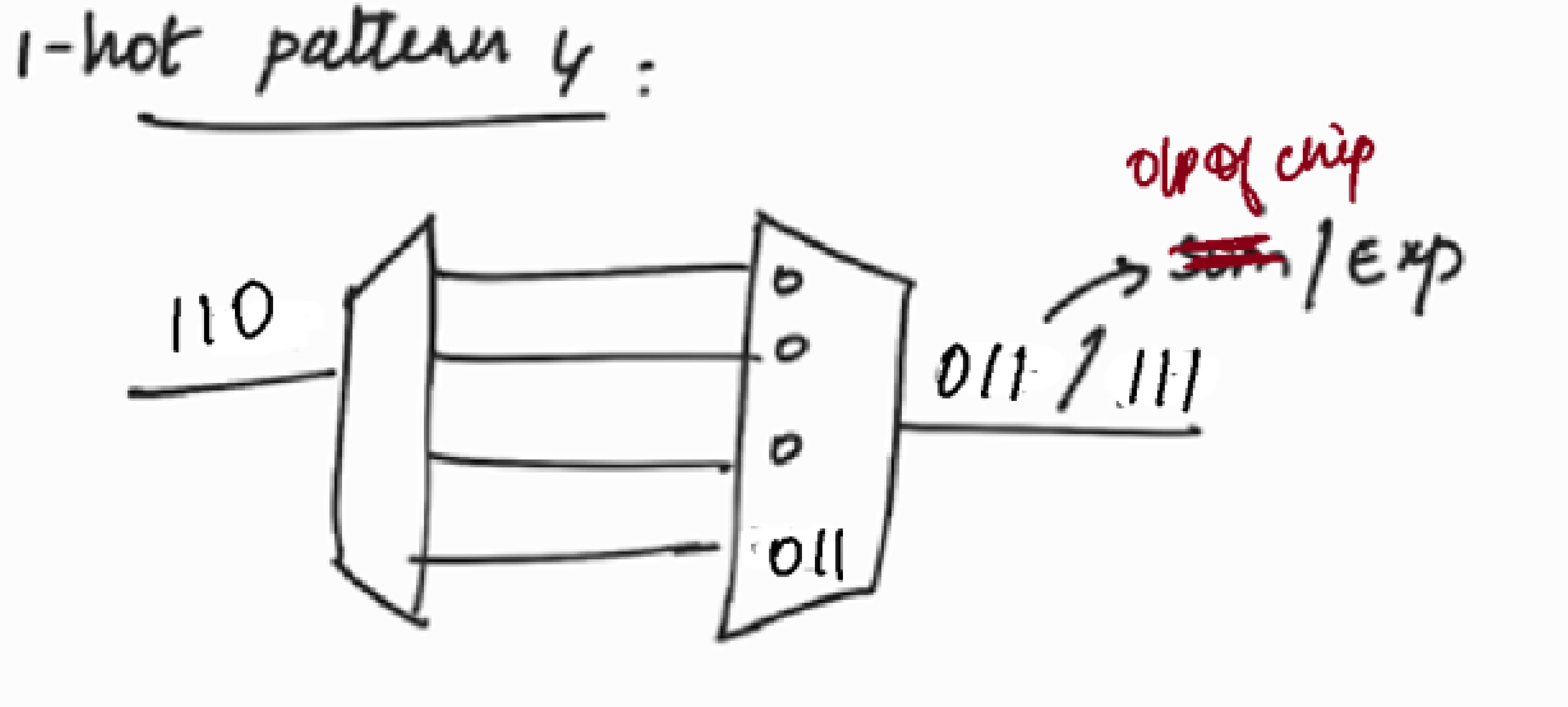
EXAMPLE

1-hot patterns









DEBUG OF EDT PATTERNS (Contd…)

1-hot patterns will observe only 1 chain at a time in a channel.
Eg: In the previous slide illustration, 1-hot pattern 2 observes only chain 2 in channel 1.

So we can find in which chain the failure is happening.

DEBUG OF EDT PATTERNS (Contd…)

The other method to debug pattern failure is to go for bypass (Bypassing the EDT logic)

Note :

We mostly go for bypass to check whether the manufacturing defect is in the EDT logic or core logic.

Bypass → pass ⟶ ~~scan~~ core logic is good
EDT logic has manufacturing defect.

→ fail ⟶ ~~scan~~ core logic has some manufacturing defect.

NOTE :-

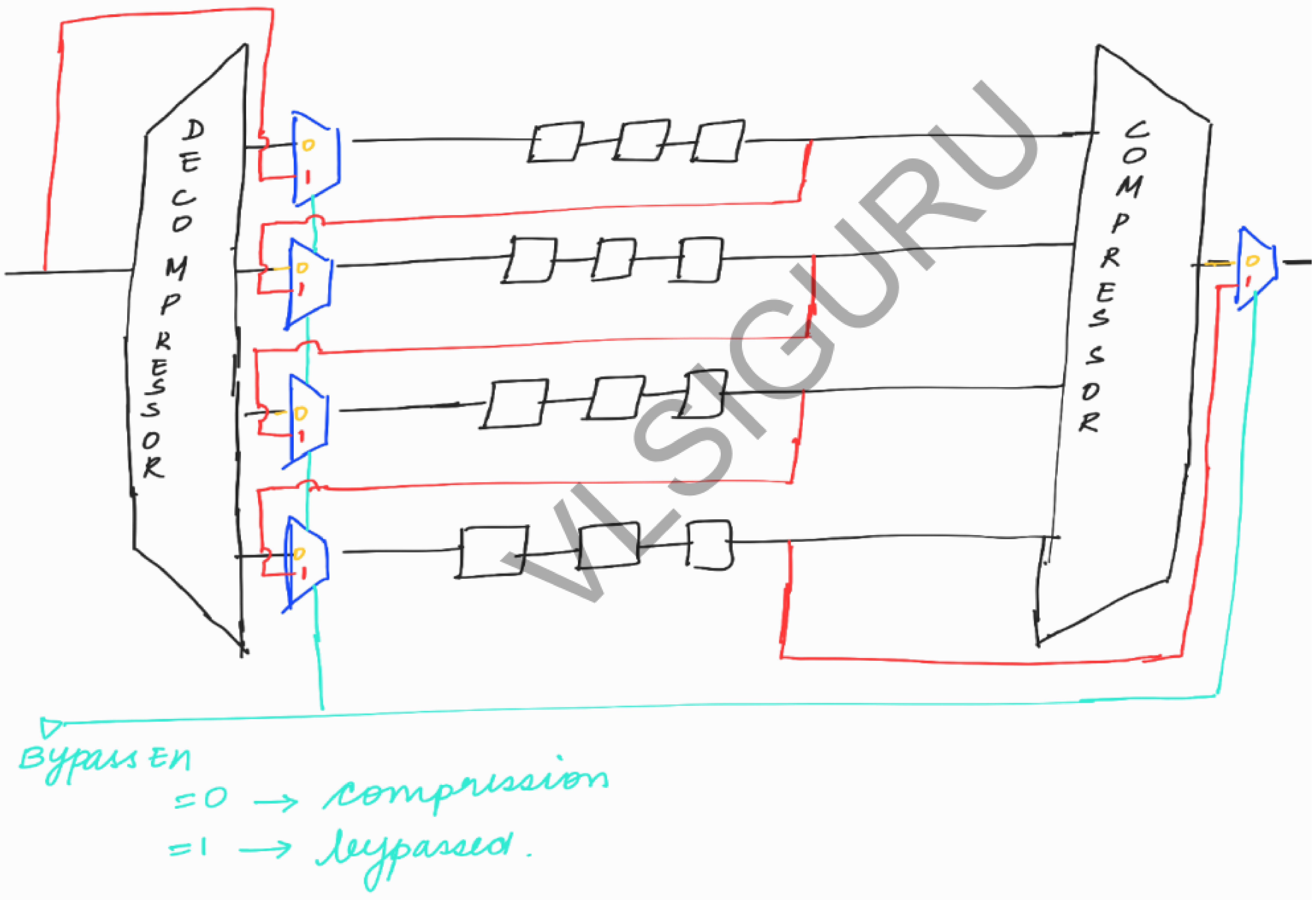
core logic = FFs + combo

EDT logic
⇒ Decomp
⇒ Comp
⇒ Controller

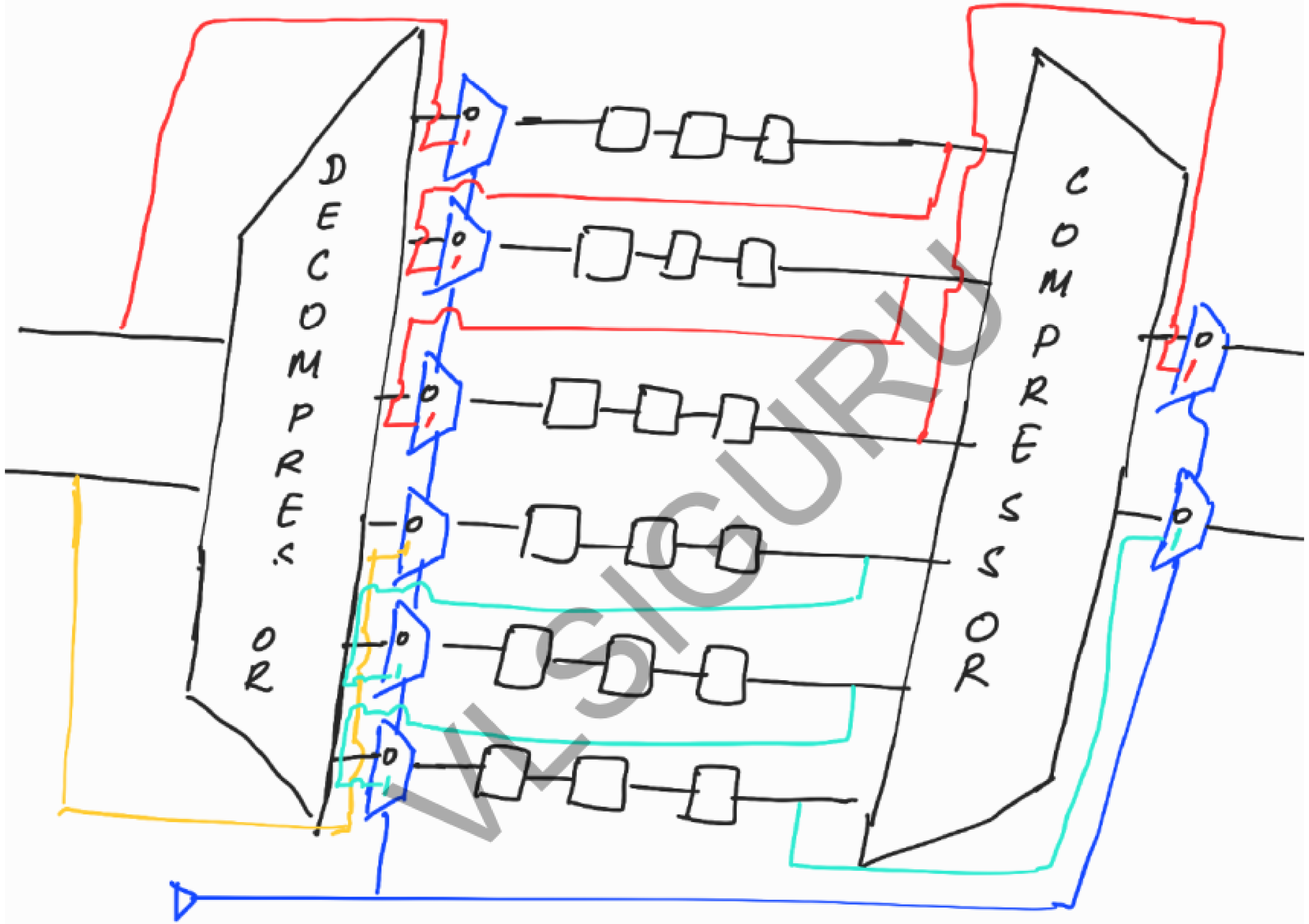
Disadvantage of Bypass : if we go for bypass, test time will increase.

DEBUG OF EDT PATTERNS (Contd…)

EDT Bypass



Bypass Example 2 :

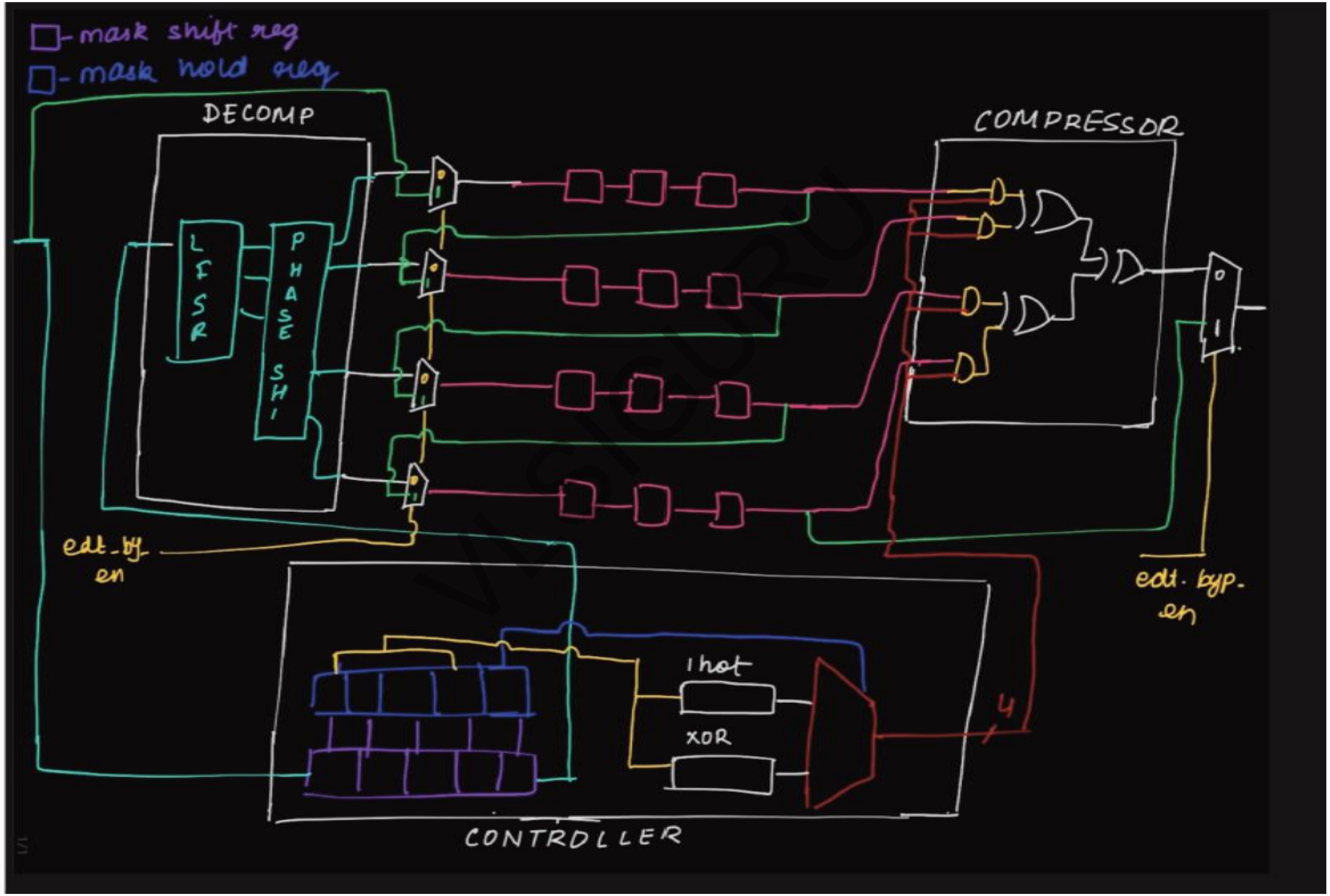


Bypass Enable

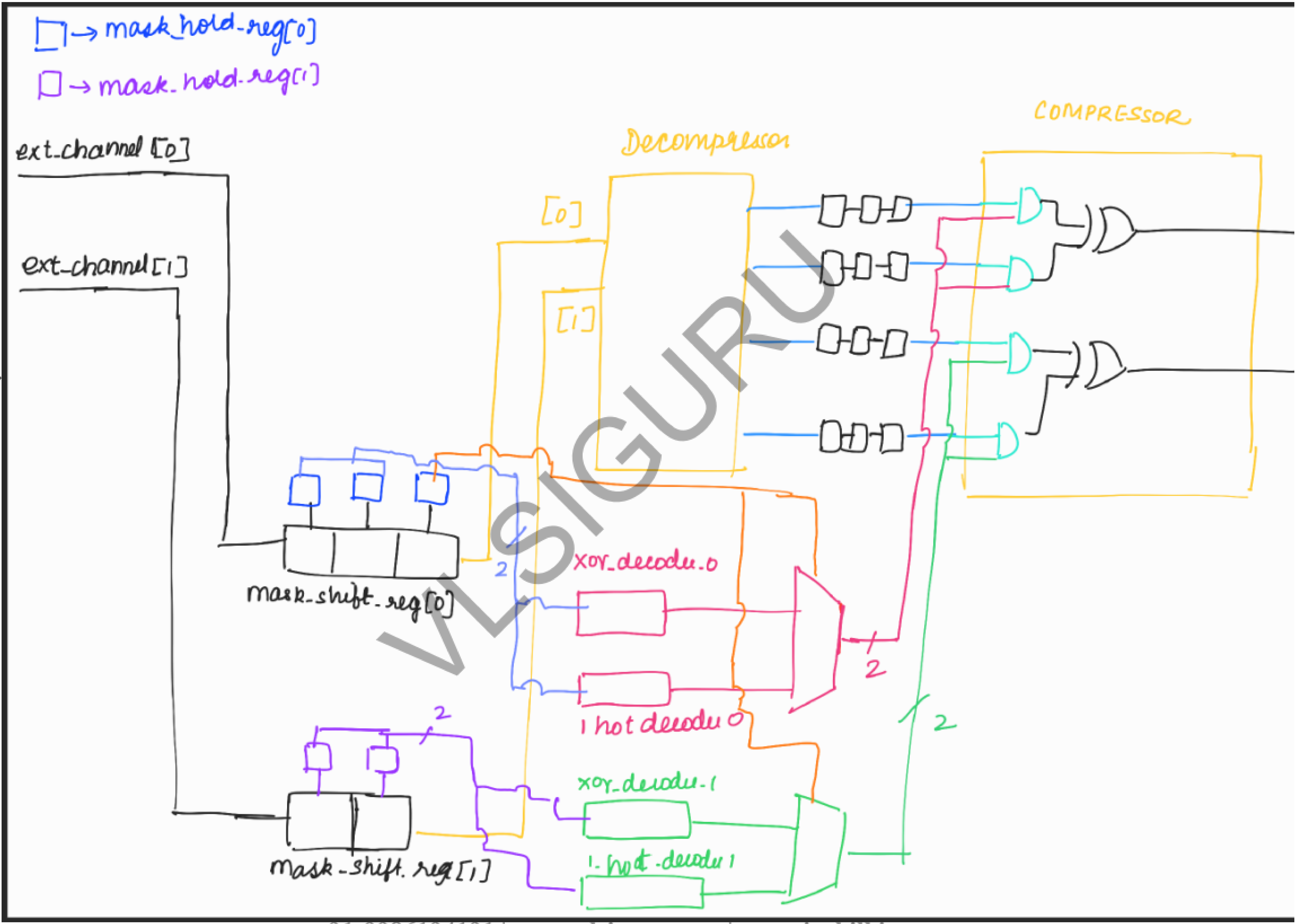
=0 with compression

=1 bypass

COMPLETE DIAGRAM



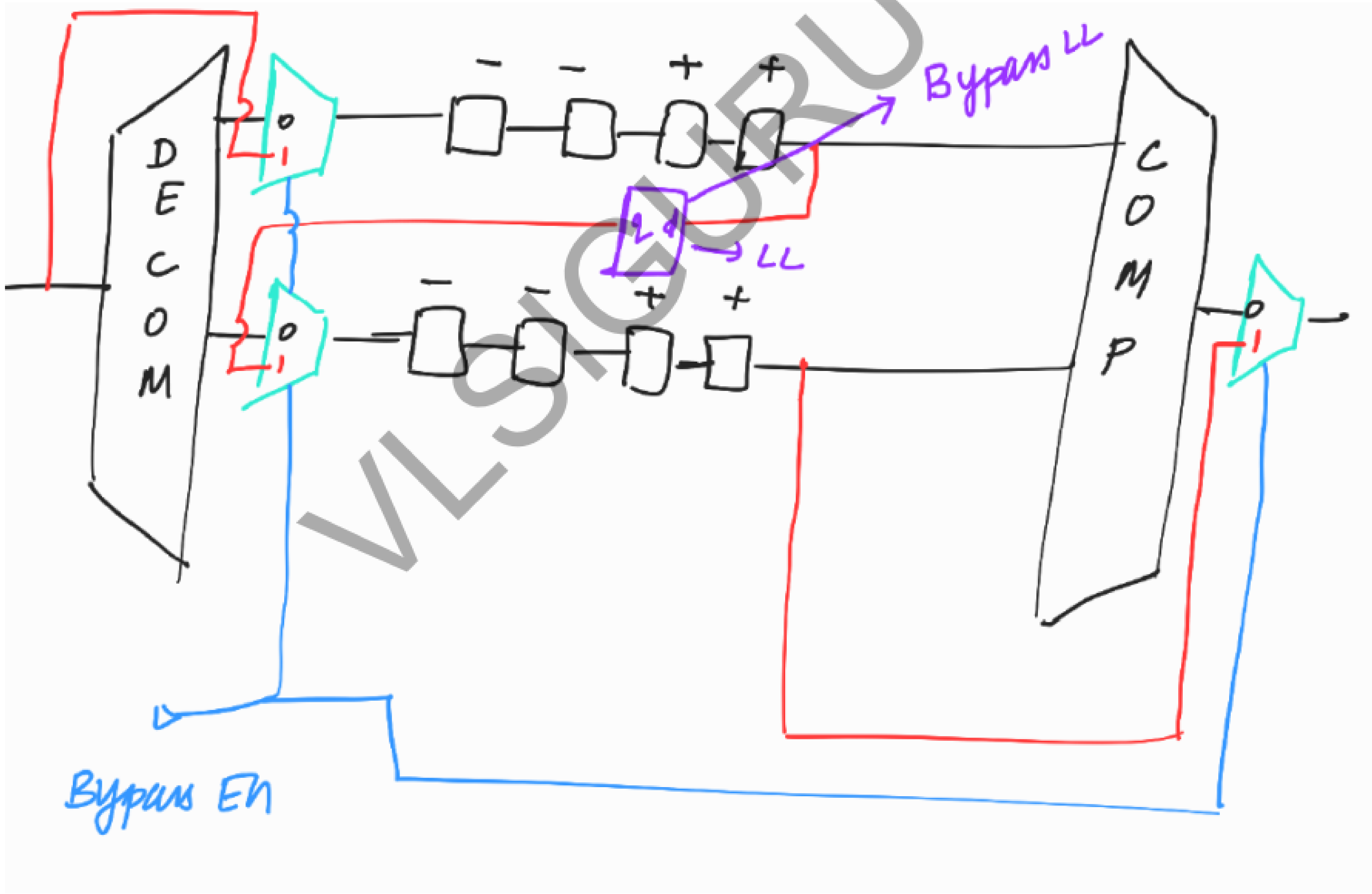
COMPLETE EDT DIAGRAM WITH 2 EXT CHANNELS



LOCK UP CELLS IN EDT

Bypass Lockup latch :

It is added whenever there is +ve flop → -ve flop during bypass



LOCK UP CELLS IN EDT (Contd…)

Lockups Between Decompressor and Scan Chain inputs:

The decompressor is located between the scan channel input pins and the scan chain inputs. It contains sequential circuitry clocked by the EDT clock. Scan chain clocking does not utilize the EDT clock. Therefore, there is a possibility of clock skew between the decompressor and the scan chain inputs.

LOCK UP CELLS IN EDT (Contd…)

Lockups at the end of scan chain (terminal lockup):

This is added because of domain mixing. The scan flop's clock is scan clk and pipeline flop's clock (inside compressor) is edt clk.

Q: What will happen if compression ratio is exceeded above 100x ?

Ans :- (i) if comp ratio is exceeded above 100x, the probability of getting fault aliasing will be more. (explanation will be given during ATPG)

(ii) XOR tree inside the compressor will be big ⇒ area ↑ses

(iii) timing violation ⇒ pipeline ffs are to be added.

Merits and De-merits of high and low compression ratio :

no. of external channel 25

10,00,000 flops in the design

Compression ratio = 50x

no. of internal chains 1250

800 flops per scan chain ==> almost equal to 800 shift cycles per pattern

no. of external channels = 25

10,00,000 flops in the design

Compression ratio = 80x

no. of internal chains = 2000

500 flops per scan chain ==> almost equal to 500 shift cycles per pattern

If we go for higher compression ratio ==> more no.of internal chains ==> Test time and Test Data Volume will be less

But if we increase compression ratio a lot (above 100x), then we have the disadvantages ==> as discussed